

A word of caution regarding ICMP

The Internet Control Message Protocol (ICMP) is a diagnostic tool and so should not be dropped outright. Any attempt to fully block ICMP is unwise as it will also stop giving error messages to us. Enable only the most important control messages such as echo-request, echo-reply, destination-unreachable and time-exceeded, and reject the rest. Echo-request and echo-reply are part of ping. In the input, we only allow echo reply and in the output, we only allow the echo-request.

```
nft add rule inet filter input iifname
enpXsY icmp type { echo-reply,
destination-unreachable, time-exceeded }
limit rate 1/second accept
nft add rule inet filter input iifname
enpXsY ip protocol icmp drop
```

Finally, we are logging and dropping all the invalid packets.

```
nft add rule inet filter input iifname
enpXsY ct state invalid log flags all
level info prefix "Invalid-Input: \"
nft add rule inet filter input iifname
enpXsY ct state invalid drop
```

Forward and output policy

In both the forward and output policies, we will drop packets by default and only accept those that are established connections.

```
nft add chain inet filter forward { type
filter hook forward priority 0 \; counter
\; policy drop \; }
nft add rule inet filter forward ct
state established, related accept
nft add rule inet filter forward ct
state invalid drop
nft add chain inet filter output { type
filter hook output priority 0 \; counter
\; policy drop \; }
```

A typical desktop user needs only Port 80 and 443 to be allowed to access the Internet. Finally, allow acceptable ICMP protocols and drop the invalid packets while logging them.

```
nft add rule inet filter output oifname
```

```
enpXsY tcp dport { 80, 443 } ct state
established accept
nft add rule inet filter output oifname
enpXsY icmp type { echo-request,
destination-unreachable, time-exceeded
} limit rate 1/second accept
nft add rule inet filter output oifname
enpXsY ip protocol icmp drop
nft add rule inet filter output oifname
enpXsY ct state invalid log flags all
level info prefix "Invalid-Output: \"
nft add rule inet filter output oifname
enpXsY ct state invalid drop
```

Now we have to save our rule-set, otherwise it will be lost when we reboot. To do so, run the following command:

```
sudo nft list ruleset. > /etc/nftables.
conf
```

We now have to load nftables at boot, for that enables the nftables service in *systemd*:

```
sudo systemctl enable nftables
```

Next, edit the nftables unit file to remove the *Execstop* option to avoid flushing the rule-set at every boot. The file is usually located in */etc/systemd/system/sysinit.target.wants/nftables.service*. Now restart the nftables:

```
sudo systemctl restart nftables
```

Logging in rsyslog

When you log the dropped packets, they go straight to syslog, which makes reading your log file quite difficult. It is better to redirect your firewall logs to a separate file. Create a directory called *nftables* in */var/log* and in it, create two files called *input.log* and *output.log* to store the input and output logs, respectively. Make sure rsyslog is installed in your system. Now go to */etc/rsyslog.d* and create a file called *nftables.conf* with the following contents:

```
:msg,regex,"Invalid-Input: " -/var/log/
nftables/Input.log
:msg,regex,"Invalid-Output: " -/var/log/
nftables/Output.log
& stop
```

Now we have to make sure the log is manageable. For that, create another file in */etc/logrotate.d* called *nftables* with the following code:

```
/var/log/nftables/* { rotate 5 daily
maxsize 50M missingok notifempty
delaycompress compress postrotate invoke-
rc.d rsyslog rotate > /dev/null endsript
}
```

Restart nftables. You can now check your rule-set. If you feel typing each command in the terminal is bothersome, you can use a script to load the nftables firewall. I hope this article is useful in protecting your system. **END** 

References

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